

Upper Division CSE Technical Electives (TE's) Historically offered: **Fall offered course** **Spring offered course**
 CSE TE's (5 Courses required - Only Upper Division CSE courses) CSE 1**='s CSE Upper Division Course
 (Excluding CSE 100 and CSE 120 Core requirements from TE's) **AND** CSE Additional Engineering TE's (3 courses
 required) Select from: CSE 1**, Engr 1**, Engr 057, Math 131 or Math 141. Courses may not Double count in the
 TE and Add CSE Engr TE. 8 total Technical Electives (TE's) required to graduate. Select TE's for your specific
 career planning.

- Technical Elective Requirements: [Requisite Override Enrollment Form](#)
 (<https://registrar.ucmerced.edu/forms>) This add form is used to obtain instructor signature if a student
 doesn't meet the designated requisite(s) for a course. Requisites include course prerequisites or co-
 requisites and major or class level restrictions.
- Major Core Requirements: [Petition for Program Variance \(PPV\)](#) (<https://enr-advising.ucmerced.edu/forms>)
 Students may submit the online [Petition for Program Variance](#) to request a non-approved course as an CSE
 Technical Elective. Note – Faculty approval does not guarantee approval of the Petition for Program Variance.
 Petition review time is approximately 6+ weeks after submission.

Note - Course term offerings subject to change. You will be required to meet any Faculty updated
 Prerequisites. Review the Course Schedule for term course offerings and updated prerequisites [https://reg-
 prod.ec.ucmerced.edu/StudentRegistrationSsb/ssb/term/termSelection?mode=search](https://reg-prod.ec.ucmerced.edu/StudentRegistrationSsb/ssb/term/termSelection?mode=search)
 *CSE 165 was discontinued after S24. CSE TE prerequisites will be updated to reflect the change. Review the
 Course Schedule for updated prerequisites.

Courses	Units	Title	Prerequisites
CSE 106 Offered F23	4	Exploratory Computing - Our ability to manipulate data depends on and is limited by our familiarity with computing technologies. We study tools for exploratory computing, emphasizing programming and scripting languages over point-and-click interfaces. We cover the Unix basics and common utilities, regular expressions, Perl and R languages. Development of a problem-solving ability to learn languages independently and cull online documentation.	CSE 031, CSE 100, Math 024. JR & SR
CSE 107 Offered F23	4	Intro to Digital Image Process - Overview of digital image processing including visual perception, image formation, spatial transformations, image enhancement, color image representations and processing, edge detection, image segmentation, and morphological image processing.	CSE 031, CSE 100, Math 024, SO, JR & SR
CSE 108 Offered S24, F24	4	Full Stack Web Development - Explores topics related to both front end and backend web development including web security, scalable architecture, web frameworks, databases, and object relational mappers.	CSE 031 and Math 024; CSE 100 (concurrent ok)
CSE 111 Offered F23, F24	4	Database Systems - Principles of database design and operation. Relational data model. High-level data modeling representations. SQL database language. Active databases with constraints and triggers. Query optimization with views and indexes. Exploiting database servers within programming languages for web application development. Other topics include transaction processing and recovery, user-defined functions, and data warehousing.	CSE 031, CSE 100, Math 024. SO, JR & SR
CSE 126/ MGMT 126 (historically not offered)	4	Information Systems and Service Design - Presents an end-to-end view of the design life cycle for information systems and services. Explains how design problems are conceived, researched, analyzed and resolved in different types of organizations and contexts, including start-ups, enterprises with legacy-systems, non-profit and government entities.	MBE & CSE majors only

CSE 130 (historically not offered)	4	Introduction to Cryptography - Introduces the basic theory and practice of cryptographic techniques used in computer security by exploring the inner workings, mathematical assumptions and foundations of cryptographic primitives and how to use them correctly. Topics include pseudorandom functions, one-way functions, cryptographic hashing, encryption (symmetric and public-key), message integrity, message authentication codes, digital signatures, key management, security protocols and more. Covers the necessary mathematical background, formal definitions of security and proofs of security based on precise definitions and assumptions.	CSE 031, CSE 100, Math 024. A strong mathematical background in probability and statistics is preferred.
CSE 135 (historically not offered)	4	Introduction to Theory of Computation - Understanding the inherent capabilities and limitations of computers is a fundamental question in computer science. To answer this question, we will define formal mathematical models of computation and study their relationships with formal languages. Topics will consist of three central areas of the theory of computation: automata, computability, and complexity.	CSE 031, CSE 100, Math 024. JR & SR
CSE 140 Offered S24	4	Computer Architecture - Fundamental concepts of digital computer design, including instruction sets, memory systems and registers, logic and mathematics units, modern CPUs and their architectural features are discussed along with the motivation for each design choice.	CSE 031, CSE 100, Math 024, Engr 065. JR & SR
CSE 141 Offered S24	4	Introduction to Compiler Construction - Explores basic concepts and techniques in compiler construction. Examines topics such as, lexical analysis, finite automata, parsing, context-sensitive analysis, intermediate representation, code shape analysis, code optimizations, code generation, and register allocation.	CSE 031, Math 024, CSE 100 Major restriction 11/13/23
CSE 150 Offered F23	4	Operating Systems - Concepts of computer operating systems including concurrency, memory management, file systems, multitasking, performance analysis, and security. Crossroads	CSE 031, CSE 100, Math 024. JR & SR
CSE 151 (historically not offered)	4	Advanced Operating Systems - Introduces the concept of advanced Operating Systems, including distributed operating system, real-time operating systems, advanced computer networks and security. Covers both fundamental knowledge and top-flight research topics in Operating Systems.	CSE 150. Conjoined w/ EECS 251. JR & SR
CSE 155 Offered S24, F24	4	Intro. to Human-Computer Interaction - Introduces students to the basic concepts in the theory and practice of Human-Computer Interaction (HCI). Teaches how hardware and software design influence the interaction between humans and computers to provide insights into the design and development of safe, effective, and enjoyable interactive systems. Crossroads OR Writing in the Discipline (WID Only through SP24.) Ethics, Scientific	CSE 031, CSE 100, Math 024. JR & SR
CSE 157 (historically not offered)	4	Signal Processing for Internet of Things - Introduces methods for signal processing between digitization and inference algorithms, where time-series signals are pre-processed. Explores the applications of Internet of Things (IoT) systems, smart connected devices in people's lives, and examples of converting physical world analog signals to digital ones. Includes topics both theoretical and practical.	CSE 031, CSE 100, Math 024.
CSE 160 Offered F23, F24	4	Computer Networks - Design concepts and implementation features of computer networks. Concepts of network robustness, scalability, addressing, routing, and security. Several contemporary networking protocols are analyzed. Recommended to take in your last year. Crossroads. Ethics, Sustainability OR Media (Media not applicable for CY2324)	CSE 031, CSE 100, Math 024. JR & SR
CSE 161 (historically not offered)	4	Wireless Networks	Not in CY23/24
CSE 162 Offered F23, F24	4	Mobile Computing - Introduces mobile cloud computing's basic concepts, including different types of mobile computing devices and applications, the communication technologies used in modern smartphones, the context-aware computing technologies, and programming on mobile devices.	CSE 031, CSE 100, Math 024, CSE JR & SR
CSE 165 Offered F23/S24	4	Intro to Object Orient Program - Topics include object-oriented programming concepts, such as classes, objects, methods, interfaces, packages, inheritance, encapsulation, and polymorphism. DISCONTINUED S24	CSE 031, CSE 100, Math 024. JR & SR

CSE 168 Offered F23	4	Distributed Software Systems - Covers the foundations of distributed software systems and their practical designs. Includes concepts and principles of distributed systems (e.g., consistency, availability, scalability, etc.); programming models for distributed computing (e.g., batch/streaming processing, RPC, Key-Value, etc.); distributed storage systems (e.g., indexing, replication, partition, and erasure coding, etc.); and convergence of HPC, Big Data, AI, and Cloud Computing with modern distributed systems. Scientific.	CSE 031, CSE 100, Math 024. JR & SR
CSE 170 Offered S24	4	Computer Graphics - Basic algorithms in computer graphics enabling students to understand and experience the process of implementing modern computer graphics applications. Topics covered: programmable shaders, rasterization, hidden surface removal, transformations, rendering pipeline, scene graphs, curves and surfaces, boundary representation, spatial partition methods, keyframe animation, texture mapping, illumination and shading.	CSE 031, CSE 100, Math 024, (CSE 165 OR Engr 140). JR & SR
CSE 173/ Cogs 123 Offered S24	4	Computational Cog Neuroscience - Design and analysis of computational simulations of human behavior and brain function. Techniques for modeling active membranes, individual neurons, the dynamics produced by recurrent excitation and lateral inhibition, synaptic plasticity, and the computational role of neurotransmitters. Formal models of perception, attention, learning, memory, language, categorization, and cognitive control.	Cogs 001 & Upper Div Cogs
CSE 175 Offered F23, F24	4	Intro to Artificial Intelligence - Overview of the main concepts and methods underlying the construction and analysis of intelligent systems, including agent architectures, problem solving, heuristic search, knowledge representation, reasoning, planning, communication, perception, robotics, and machine learning. Includes a laboratory component in which intelligent systems are constructed and examined. Crossroads. CY24 No Longer Cross listed with Cogs 175.	CY21: CSE 019 OR CSE 021 (Cogs 001 recommended) CY22/23: CSE 019 OR CSE 021 OR CSE 022 (Cogs 001 recommended)
CSE 176 Offered F23, F24	4	Intro to Machine Learning - Survey of techniques for development and analysis of software that learns from experience. Specific topics: supervised learning (classification, regression); unsupervised learning (density estimation, clustering, dimensionality reduction); reinforcement learning; and others. Specific techniques: linear classifiers, mixture models, nonparametric methods, decision trees, neural networks, kernel machines, ensembles, graphical models, Bayesian methods, etc. Crossroads. CY22&23&24: Scientific.	CSE 031, CSE 100, Math 141. JR & SR
CSE 177 (historically not offered)	4	Database Systems Implementation - Studies the internals of a database management system, with emphasis on query execution. The final goal of the class is to build a fully functional database execution engine consisting of all the standard components: storage manager, buffer manager, query execution engine, query optimizer, and query compiler.	CSE 031, CSE 100, Math 024, CSE 111. JR & SR Conjoined EECS 277
CSE 178 (historically not offered)	4	Computer Network Security - Introduces fundamental concepts in the design and development of secure computer networks. Covers security threats, secret-key and public-key cryptography and algorithms, digital signatures, authentication, Electronic mail, Public-key infrastructure, viruses and worms. Crossroads.	CSE 031, CSE 100, Math 024, CSE 150. JR & SR
CSE 179 (historically not offered)	4	Intro to Parallel Computing - Parallel computing is pervasive. From embedded devices, laptops, to high-end supercomputers, and large-scale data centers, parallel computing is widely employed to achieve performance and efficiency targets. This course introduces the foundations of parallel computing, including parallel architectures, parallel programming methods and techniques, and parallel algorithm designs. Crossroads.	CSE 031, CSE 100, Math 024, CSE 140. JR & SR.
CSE 180 Offered S23, F23	4	Introduction to Robotics - Covers the basics of robotics focusing on the algorithmic side, rather than technology. Introduces basic computational techniques concerning spatial modeling, planning, and sensor processing. The course has a strong hands-on component. Implementation of different techniques in simulation will complement the theoretical lectures. Crossroads.	CSE 031, CSE 100, Math 024. JR & SR

CSE 185 Offered F23, S24	4	Introduction to Computer Vision - Overview of fundamental image processing and pattern recognition techniques including image formation, edge detection, image segmentation, optical flow, recovery of three-dimensional structure from shading or stereo information, shape representations, and issues in object recognition. Crossroads. CY22 & 23 Media	CSE 031, CSE 100, Math 024. JR & SR
CSE 190 (historically not offered)	1-4	Topics in Computer Science and Engineering - Covers special topics within Computer Science and Engineering that are not covered in the regularly scheduled courses. Each class will have a specific focus which will be defined by the instructor of the course on a semester-by-semester basis.	CSE 031, CSE 100, Math 024. CSE Majors
CSE 195	1-4	CSE 195: Upper Division Undergraduate Research - Independent Study/Research Form (online) https://registrar.ucmerced.edu/forms A maximum of 4 units may apply to the requirement. You would contact CSE Faculty (https://engineering.ucmerced.edu/faculty) to work on-on-one with them in their area of Research. Additional units may be taken for Pass/No Pass with the Grade Mode Form , and would be listed under Additional Coursework. https://registrar.ucmerced.edu/forms#Grade%20Forms Fall/Spring: 1 unit = 's 3 hours of Research/week Summer: 6 Week Session: 1-unit = 's 8 hours Research/week. 8 Week Session: 1-unit = 's 6 hours research/week.	

Audit #8 CSE Additional Engineering Technical Electives (CSE Additional TE's)

Courses	Units	Title	Prerequisites
Engr 057 Engineering based	4	Statics and Dynamics - Fundamentals of statics. Kinematics and equations of motion of a particle for rectilinear and curvilinear motion. Planar kinematics of rigid bodies. Kinetics for planar motion of rigid bodies, including equations of motion and principles of energy and momentum.	Math 021, Phys 008(H)/018
Engr 1**	4	ENGR 1** = 's Engineering Upper Division Course Verify prerequisites	
Engr 155 Offered F23/S24/F24	3	Engineering Economic Analysis – Microeconomic principles and methods. Time value of money, interest and equivalences, analysis of economic alternatives, depreciation, inflation, and taxes, estimates of demand, cost and risk, decision theory.	JR & SR
Engr 156 Offered F23	3	Technical Writing for Scientists & Engineers - Instructs students on writing scientific and technical documents. Specific assignments will include writing a scientific abstract versus a summary, a research or design proposal, the lab report, and an original research article. Data will be provided for graphical presentation and statistical analyses. Differences between a proposal and a journal article will also be highlighted.	Wri 010, SO, JR, SR Also Fulfills Writing in the Discipline requirement
Engr 180 Offered F23/S24/F24	4	Spatial Analysis & Modeling - Principles of geographic information systems [GIS]; applications of GIS to environmental, water, and resource management issues; problem solving with GIS. Other topics include spatial analysis interpolation techniques and model integration. Crossroads requirement, CY2122, 223, 2324: Global, Scientific, Sustainability OR Media (Media removed for CY2324) Intellectual Experiences	Math 021
Math 131 Offered F23/S24/F24	4	Numerical Methods for Scientists & Engineers - Introduces numerical methods with emphasis on algorithm construction and implementation. Programming, round-off error, solutions of equations in one variable, interpolation and polynomial approximation, approximation theory, direct solvers for linear systems, numerical differentiation and integration, initial-value problems for ordinary differential equations.	ME 021 OR CSE 019 OR CSE 022 OR (CSE 020/021) & Math 024
Math 141 Offered F23/S24/F24	4	Linear Analysis I - Applied linear analysis of finite dimensional vector spaces. Review of matrix algebra, vector spaces, orthogonality, least-squares approximations, eigenvalue problems, positive definite matrices, singular value decomposition with applications in science and engineering.	Math 023 & Math 024